

Construction phase plan

Company name: Serco Description of project: Refurbishment of Ward 3 in Forth Valley Royal Hospital's Mental Health Unit. Works comprise bedroom and en-suite refurbishment, including ligature-safe Kingsway FD30S doorsets, replacement ligature-safe windows with integral blinds, ligature-safe wardrobes, IPS replacement, Altro Whiterock hygienic wall cladding, Altro Aquarius flooring, sanitaryware removal/refit, plumbing alterations, electrical isolations, fused spur installation, Pinpoint controls work and replacement ligature-safe emergency bulkhead lights. Proposed start date: 05/05/2026 Expected duration: 15 week programme - phased room-by-room programme with 3 rooms decanted and refurbished at a time.	
1. Programme & project details, including any important dates.	Works will be undertaken in a live mental health ward (Group 2 infection risk environment) and sequenced to maintain service continuity. Proposed phasing is: ○ Pre-start: confirm room decant strategy, barriers, clean and dirty routes, permits and infection control controls. ○ Construction: isolate services, strip-out, install ligature-safe doors/windows, sanitaryware, IPS, cladding and flooring, then recommission. ○ Validation: electrical testing, plumbing flushing, temperature checks, ventilation checks, enhanced cleaning, Pinpoint and IPC sign-off prior to reoccupation, door, windows, blinds operation and review ligature risk using the NHS ligature assessment process. ○ Handover: H&S folder, O&M information, HAI-SCRIBE Stage 4 review, test records and room sign-off.
2. * Details of the project team, including:	
a) Client	NHS Forth Valley (represented by Serco FM Lifecycle Project Manager, Kevin Fotheringham).
b) Principal Designer (PD)	Currie Brown. Principal Designer to manage, monitor and coordinate the pre-construction phase and communicate residual design risks.
c) Designer(s)	Designer: (specified by NHS – must comply with HBN 03-01 and Scotland Assure) Project design contributors include specialist suppliers and installers for anti-ligature doorsets, ligature-safe windows, IPS, flooring, hygienic wall cladding, electrical and plumbing services.
d) Principal Contractor (PC)	Alan Grant (PC / Serco FM CDM Representative). Principal Contractor to plan, manage, monitor and coordinate the construction phase and review Contractors RAMS.
e) Contractor(s)	Specialist trade contractors include: ○ Windows – Polar NE. Nick Sutton (Project Manager) ○ Doors – Kingsway. Jonathan Marsden (Consultant)

Construction phase plan

	○ Rooms (& ensuite) – MPMH. Steve Welch (Contracts Manager) ○ Door sensor system – Pinpoint Craig Lamont (Account Manager)
f) Other Consultants.	○ Serco Lifecycle PM / HAI-SCRIBE - Kevin Fotheringham, ○ Ward Manager / Nurse in Charge: Leigh-Ann Colvin ○ NHS Estates: Jamie Tripney ○ NHS IPC: Diane Campbell ○ NHS Compliance Manager / HAI-SCRIBE lead: – David Rentoul ○ NHS FV Health & safety services James O’Kane ○ Fire Safety Advisor: Alan McGeachie
3. Extent and location of existing records and plans which are relevant to health and safety on site, including information on existing structures when appropriate.	Relevant pre-construction information includes HAI-SCRIBE Stage 2 & 3 assessments, refurbishment details, fire strategy information, existing drawings, M&E and water services information ventilation information, and local NHS / Serco H&S and IPC procedures. The safeguard system for any local ward 3 NHS FV Health & Safety risk.

Construction phase plan *continued*

Management of the work	
1. Management structure and responsibilities.	○ Principal Contractor to have day-to-day control of the works. ○ Serco Lifecycle PM acts as the Client's representative and coordinates with NHS Estates and IPC. ○ Ward Manager / Nurse in Charge controls patient and staff operations within the ward. ○ IPC / HAI-SCRIBE lead approves infection control arrangements and sign-off. ○ All subcontractors report through the Principal Contractor.
2. Health, safety & infection control goals for the project and arrangements for monitoring and review of health and safety performance.	Project goals are as follows: ○ Zero harm to patients, staff, visitors and contractors. ○ No uncontrolled dust or infection transmission. ○ No ligature-risk exposure arising from the works. ○ Safe room-by-room hand-back. ○ Monitoring will include daily site inspections by the PC. ○ Weekly joint inspections with Serco/NHS Estates/IPC and PC. ○ Review of permits and RAMS compliance. ○ Cleanliness checks and barrier integrity checks.
3. Health, safety & infection control arrangements for the construction phase.	All works to comply with: ○ CDM Regulations 2015, ○ HSWA 1974, ○ Management Regulations 1999, ○ COSHH, ○ PUWER, ○ NHS Forth Valley / Serco H&S and ○ NHSFVRH IPC policies, ○ SHFN 30 / HAI-SCRIBE, ○ Relevant SHTM guidance including SHTM 03-01, 04-01, 06-01 and 64. ○ HBN 03-01 (2014) - including IM/2020/012. ○ Construction areas will be separated from occupied clinical space by full-height sealed temporary partitions, controlled access and agreed clean/dirty routes. ○ Dust suppression, enhanced cleaning, water hygiene controls, permit-to-work arrangements and room sign-off procedures will be implemented throughout.
4. Site rules	○ No access by patients to work areas. ○ Site sign in/out is mandatory. ○ PPE and task-specific PPE / RPE to be worn as required by RAMS. ○ No smoking/vaping on site. Tools, cords, straps, sharps and other ligature-risk items to be removed or locked away when not in immediate use. ○ Barriers, fire routes and clinical routes must not be compromised. ○ Good housekeeping and hand hygiene to be maintained at all times.

Construction phase plan *continued*

5. Arrangements for:	
a) Co-operation between the project team on site and co-ordination of their work	○ See Communication Plan full details ○ Weekly coordination meetings involving Principal Contractor, Serco, NHS Estates, IPC and Ward representatives. ○ Daily start-of-shift briefings and permit reviews. This is likely to cross ward: day/back/night shift patterns. ○ Sequencing agreed room by room. ○ A lessons-learned will be carried out at the end of each phase to address any issues experienced ○ In addition, room allocation for the next phase will be agreed with the clinical team. ○ Subcontractors will be coordinated by the Principal Contractor.
b) Consultation with the workforce	○ Consultation through inductions, daily briefings - which must be attended by the local clinical lead at the time, toolbox talks and supervisor engagement. ○ Shift handovers for clinical teams. ○ Hot / Cold debriefs after any adverse events. ○ Workforce encouraged to report concerns, accidents / adverse events, near misses and suggested improvements immediately.
c) The exchange of design information between the client, designers, principal designers and contractors on site	Design and residual risk information to be issued by the Principal Designer / design team to the Principal Contractor and briefed to supervisors, local NHS management team (if appropriate) and subcontractors before work starts or changes are introduced.
d) Handling design changes during the project	Any design change affecting the programme must, in the first instance, be raised using the Change Request System before being reviewed by the Principal Designer, Client representative, NHS Estates, local ward management team (if appropriate) and IPC before implementation.
e) The selection and control of contractors	Contractors and subcontractors to be selected on proven competency, healthcare experience, training, supervision standards, insurance, and ability to work to HAI-SCRIBE and anti-ligature requirements.
f) The exchange of health and safety information between contractors	Relevant H&S information to be exchanged through pre-start meetings, CPP, RAMS, permits, coordination meetings, daily briefings and close-out / handover records.
g) Site security	Controlled access to ward work zones by sign-in/out and badge system. Materials and tools stored in locked areas and a method of counting in / out the tools in place and removing them from the site after use at the end of each shift

Construction phase plan *continued*

	Deliveries and waste movements controlled to be always supervised by staff to avoid unauthorised access and patient contact.
h) Site induction	All personnel to attend a Ward 3-specific induction covering live hospital working, mental health / behavioural considerations, anti-ligature awareness, IPC controls, emergency arrangements, fire procedures, welfare, reporting requirements and the current RAMS and specific site rules.
i) On-site training	Toolbox talks and task briefings to cover dust control, water hygiene and flushing, permit systems, manual handling, safe use of access equipment, behavioural awareness in the mental health setting, and manufacturer requirements for specialist anti-ligature products.
j) Welfare facilities and first aid	Welfare facilities to be located away from clinical spaces. No construction PPE or contaminated clothing to be worn in patient areas. Hand hygiene required when entering/leaving ward areas. Principal Contractor to provide first aid arrangements and ensure trained first aiders are available.

Management of the work <i>(continued)</i>	
k) The reporting and investigation of accident and incidents including near misses or those involving patients who have accessed unattended areas / tools under the control of contractors	All accidents, incidents and near misses to be reported immediately to the Principal Contractor and recorded. Patient-related incidents or concerns to be escalated immediately to the Ward Manager / Nurse in Charge and Serco representative. HAI incidents to trigger stop work, containment and IPC review.
l) The production and approval of risk assessments and written systems of work	Task-specific RAMS and written systems of work required for all significant activities including strip-out, doors/windows replacement, flooring and cladding works, plumbing alterations, electrical isolations, water hygiene controls, access equipment, waste removal and cleaning. RAMS to be approved by the Principal Contractor before works start, in addition specific RAMS required for potential work tasks that could create foreseeable risks for patients / NHS staff
m) Fire safety procedures	Hospital fire procedures to apply. Hot works only under permit. Temporary partitions to maintain fire compartmentation as required. Escape routes, alarm points and firefighting equipment to remain accessible. Fire watch and extinguishers to be provided for hot works.
n) Emergency precautions and procedures	Emergency procedures to align with hospital arrangements. In fire or medical emergency, comply with ward instructions and hospital alarm procedures. Emergency (PIT) alarms to be used in line with procedures provided. Suspected infection control breach, dust escape or water quality issue requires immediate stop work, make safe and IPC review.
Arrangements for controlling significant site risks	

Construction phase plan *continued*

1. Safety risks, including:	
a) Delivery and removal of materials (including waste) and work equipment taking account of any risks to the public (for example during access to or egress from the site)	○ Deliveries via agreed clean routes avoiding patient/public routes where practicable. ○ Waste and removed materials via designated dirty routes using sealed, double-bagged or enclosed containers as appropriate to be moved in locked waste containers only. Movements to be timed to minimise interaction with ward activity and the public.
b) Dealing with services - water, electricity and gas, including overhead powerlines and temporary electrical installations	All electrical and water services to be identified and isolated in coordination with NHS Estates and permit systems before work begins. Lock-out/tag-out to apply for circuits - any wiring should not be left accessible to patients outwith patient rooms. Water systems to be managed to avoid dead legs; temporarily isolated branches to be flushed weekly with records retained.
c) Accommodating adjacent land use	Adjacent live ward operations and public hospital circulation must be protected by robust segregation, communication and planned timing of noisy, dusty or disruptive work and where necessary clinical observations of patients. Clinical operations take priority.
d) Stability of structures whilst carrying out construction work, including temporary structures and existing unstable structures	No major structural alterations are currently anticipated. Any opening-up that could affect stability or fire compartmentation will be reviewed by the design team before proceeding, with temporary support where required.
e) Preventing falls	Window replacement or similar work at height to use suitable access equipment such as MEWPs or tower scaffolds where required, with trained operatives, inspection, exclusion zones and edge protection as necessary.
f) Work with or near fragile materials	Not anticipated as a principal risk; however, any glazing or fragile surfaces encountered will be identified during task planning and protected accordingly. Should there be any broken glass or other surface to then provide a sharp edge, there must be effective clean up to remove all / any such items immediately.
g) Work involving the assembly or dismantling of heavy, prefabricated components	Doorsets, windows and other heavy prefabricated components to be handled under specific lifting / manual handling assessments using sufficient labour and mechanical aids where practicable.
h) Work near high-voltage cables	Not anticipated.
i) Work on excavations and work where there are poor ground conditions	Not applicable.
j) Work on wells, underground	Not applicable.

Construction phase plan *continued*

earthworks and tunnels	
k) Work exposing workers to the risk of drowning	Not applicable.

Arrangements for controlling significant site risks (<i>continued</i>)	
l) Work carried out by divers having a system of air supply	Not applicable.
m) Work in a caisson with a compressed air system	Not applicable.
n) Work involving explosives	Not applicable.
o) Traffic routes and segregation of vehicles and pedestrians	Traffic routes - these will change depending on the rooms/phase being worked in at specific times of the project - it is essential to discuss / communicate with clinical teams during each phase. Pedestrian segregation to be maintained between contractors, staff, patients and visitors. Delivery / collection times to be controlled and service routes used where possible.
p) Storage of materials (<i>particularly hazardous materials</i>) and work equipment	Materials and equipment to be stored only in designated secured areas. Hazardous products to be stored in accordance with COSHH assessments and manufacturer guidance. Combustible load to be minimised.
q) Work which puts workers at risk from chemical and biological substances	Operatives may be exposed to chemical substances such as adhesives, sealants and cleaning agents and to biological risks associated with live hospital working. If contractors come in contact with blood / body fluids adjacent to work areas, ensure appropriate contact with clinical staff for their processes to be applied. Controls include COSHH assessments, hygiene arrangements, appropriate PPE/RPE, segregation, cleaning and supervision.
r) Work with ionizing radiation requiring the designation of controlled or supplied areas.	Not applicable.
2. Health risks, including:	
a) The removal of asbestos	Not applicable.
b) Dealing with contaminated land	Not applicable.
c) Manual handling	Should be conducted in line with latest revision of the Manual Handling Operations Regulations 1992, and risks will be mitigated by Contractors RAMS

Construction phase plan *continued*

d) Use of hazardous substances, particularly where there is a need for health monitoring	Not applicable.
--	-----------------

Arrangements for controlling significant site risks (*continued*)

e) Reducing noise and/or vibration	Noisy activities to be restricted to agreed time windows with the Ward Manager. Vibration-producing tasks to be minimised and controlled to avoid disturbance to vulnerable patients, or patients with heightened agitation levels.
f) Exposure to UV radiation (<i>from the sun</i>)	Low risk due to predominantly internal works. Any external work will require suitable sun protection and welfare arrangements.
g) Any other significant health risks.	Significant additional health risks include dust exposure, contamination transfer, legionella / pseudomonas risk from water systems, slips/trips in constrained live areas, stress / behavioural risks in a mental health unit and sharps / ligature-risk items if poorly controlled. Controls include sealed barriers, extraction, tack mats, enhanced cleaning, flushing and temperature checks, strict housekeeping, controlled access and behavioural awareness training.

The health and safety file

1. Layout and format.	Health and safety file to be produced in an electronic format, structured by asset / room / discipline, with clear sections for residual risks, product data, test records, certificates, O&M information and cleaning / maintenance requirements.
2. Arrangements for the collection and gathering of information.	Information to be collected progressively from designers, specialist suppliers and contractors throughout the project, including drawings, specifications, product literature, certificates, commissioning data, testing records, warranties and residual risk information.
3. Storage of information.	File to be stored electronically by the Client / NHS Estates in the agreed asset information system or document repository at project completion.

Significant design and construction hazards

Construction phase plan *continued*

1. Significant design assumptions and suggested work methods, sequences or other control measures.	Key design assumptions and controls include full decant of rooms before work starts; room-by-room sequencing; working behind locked doors for refurbishment; Door replacement will take place, one at a time each day, with supervision from clinical/Serco staff during that replacement; no uncontrolled patient access to work zones; ligature-safe replacements to be installed in accordance with manufacturer details; and no reoccupation until services are recommissioned, cleaning completed and IPC sign-off obtained.
2. Arrangements for co-ordination of ongoing design work and handling design changes.	Ongoing design coordination and changes to be managed through the Principal Designer and project meetings, with all relevant parties informed of changes affecting safety, infection control, fire, services or anti-ligature performance before work proceeds.
3. Information on significant risks identified during design.	Significant risks identified during design include working in a live mental health ward, dust and airborne particulates, temporary openings during door/window replacement, infection transmission, water hygiene risks, behavioural/patient interface risks, electrical isolation, fire compartmentation and safe installation of specialist anti-ligature products.
4. Materials requiring particular precautions or any item that could be used from a violence perspective.	Materials requiring particular precautions include adhesives, sealants, flooring and cladding products (COSHH-controlled), any asbestos-containing materials if identified, sanitary fittings and products requiring strict compliance with SHTM / manufacturer guidance, and any items that could present ligature or self-harm risk, or any item that could be used from a violence perspective if left unsecured.

Comments

Monitoring and audits

Before entering ward or clinical areas, all personnel must have PVG clearance, attend site induction, clinical inductions and participate in pre-start briefings with clinical staff. Daily toolbox talks and inspections will be carried out by the Principal Contractor Site Manager. Weekly H&S and IPC audits will be conducted by a member of the Serco Projects Team. Weekly coordination meetings

Site segregation and security

Rapid door replacement during which clinical observation of patient will be always carried out on a 1:1 basis to ensure no patient access to work etc, Serco will also provide MAs to ensure contractor and patient safety. Doors and windows in work zones will be sealed during works to prevent dust and contamination spread. Tools and materials will be secured by locked toolboxes and locked rooms to prevent patient access and minimise ligature risk. All room refurbishments will be carried out behind locked doors.

Incident reporting procedures

All accidents, incidents and near misses will be reported immediately to the Principal Contractor Site Manager and recorded within the Serco incident reporting system. Incidents involving patients will be escalated to the Ward Manager, NHS Estates and Infection Prevention & Control teams where appropriate.

Fire safety controls

Construction phase plan *continued*

Hot works will be controlled through the hospital permit-to-work system. Fire extinguishers will be available in work areas. Escape routes will remain clear and hospital fire procedures will apply at all times.

Waste management

Deliveries will follow agreed clean routes while waste will be removed via designated dirty routes. Construction waste will be contained within sealed containers, locked and removed at agreed times to minimise contact with patients and staff.

Health and safety file

The Health and Safety File will be prepared by the Principal Designer and will include as-built drawings, product information, maintenance requirements, commissioning certificates and water hygiene records. The file will be handed to NHS Forth Valley Estates on completion of the project.